Alpha↔Omega Dynamics (AΩD)

The Hidden Temporal Dynamics of Stokes

The Art Of The Leprechaun: Fractal Calculus – $\mathfrak{S}$

43 °c

Manual Volume II — Matter I: Elements 1–118

Relational reference, isotope catalog and declared survival application

Reginald Poyau

November 2026

$$A \leftrightarrow_\mu \Omega \longleftrightarrow \Omega \leftrightarrow_\mu A$$

Epitaph

$$\mathcal{N} \equiv \infty \qquad \mathcal{N} \to \mathcal{N} \to \mathcal{N} \equiv \mathcal{N}$$

Comprehension, cooperation, and presence are not prerequisites of AFC.

Nuff said.

Contents

1 Manual Volume II Scope and Record Discipline . . . 3
1.1 Reading route and evidence scope . . . 3
1.2 Exact arithmetic and external-comparison quarantine . . . 3
2 Imported Relations and Conditional Support Arithmetic . . . 4
2.1 Distinction, motif and instance evidence . . . 4
2.2 Support polynomial and the reflection-duration domain . . . 4
3 Enclosure Membership and the Visual Model . . . 5
3.1 Even enclosure sequence and cumulative boundaries . . . 5
3.2 Relations before layout . . . 6
4 Conditional Support Expressions and Assigned Inputs . . . 8
4.1 Inherited fusion domain and the present calculation . . . 8
4.2 Assigned classifications, residuals and occupancy . . . 9
4.3 Complete reference ledger . . . 9
5 Verification Scope and Pending Native Evidence . . . 10
5.1 Executed static checks and inherited audit obligations . . . 10
6 Folded Construction and Relational Time . . . 11
6.1 Distinction, handed hinge and retained lineage . . . 11
6.2 An injective two-face example . . . 11
6.3 Declared recurrence and mapped comparison . . . 12
6.4 Causal returned-current response and exposure . . . 13
7 Staged Hyperfine Application . . . 14
7.1 Atomic identity and sourced recurrence . . . 14
7.2 A common model and its remaining observation map . . . 15
8 State-Dependent Evolution and the SI Prediction Experiment . . . 16
8.1 A response that changes later events . . . 16
8.2 The atomic observable is an additional construction . . . 17
8.3 Elapsed seconds and a joint uncertainty budget . . . 18
8.4 Published frequency covariance . . . 19
9 Isotope Half-Lives and a Common Survival Experiment . . . 20
9.1 Source records and nucleon bookkeeping . . . 20
9.2 A state-dependent finite law . . . 22
9.3 A measured test of the restricted specialization . . . 24
10 Reference, History and Pending-Comparison Registry . . . 24
A Exact Arithmetic Fixtures . . . 26
A.1 Scope . . . 26
A.2 Four-step decompositions . . . 26
A.3 Labelled-reference occupancy . . . 26

A.4 Falsification 27
B Pending External Comparison 27
B.1 Scope and numeric policy 27
B.2 Pending comparison gates 27
C Exact Parity, Role and Occupancy Comparators 27
C.1 Integer comparators 27
C.2 Signed-magnitude record 28
D Complete Element Reference Ledger 28

1 Manual Volume II Scope and Record Discipline

This volume supplies a compact relational reference for **Matter I: Elements 1–118**, with eight unfilled reference slots completing the declared grouping through slot 126, and a separately scoped folding and recurrence application. It applies the exact reviewed AOD M3 candidate at commit `b78b48d`, downstream of the v41.02 foundation, and the bound AFC v2.4 reference. M3 is an unmerged candidate; PR 43 is not a version 43 release. Main owns the ontology, hypotheses and mathematics; this Manual owns the elemental inputs, worked constructions, declared update model, visual model, validation and historical crosswalk.

1.1 Reading route and evidence scope

The dependency order is axiom, distinction, induced relations, admitted motif, declared model, mathematical consequence. The inherited route is Main §1, Main §2, Main §3, Main §4, and Main §5. The present visual model projects declared enclosure membership and local roles. It does not instantiate the native construction trace for each element.

The active element table records exact grouping and support arithmetic, assigned fixture inputs and their consequences. Its admission status is pending and its native execution status is `not_executed`. A support declaration is formal syntax; it is not a cycle or fusion certificate. Element-wide native D.E.C. rows and return witnesses remain pending as specified in §5. The finite construction in §6 has its own event, scope-map and declared-model records; its checks do not promote the element table's status.

1.2 Exact arithmetic and external-comparison quarantine

The support polynomial, cumulative boundaries, local counts, conditional support sums and assigned-input residuals use exact integers. Labelled-slot occupancy and the declared model's temporal comparisons use exact rationals. Arithmetic exactness, structural admission, declared-model execution, metrology and empirical agreement are separate claims. The hydrogen/caesium source inventory in §7 supplies external state identities and a retrospective target. §8 adds sequential state dependence, the SI measurement layer, a conventional benchmark and an executable underdetermination test for the atomic observation law. The AOD atomic point prediction remains absent. §9 adds a source-qualified isotope catalog and an exact survival ensemble whose reference-return count is distinct from SI elapsed time. App. B records the comparison boundary.

Provenance. The paired Main publication is built from the pinned M3 candidate source before this Manual. The earlier v41.02 source and source-dependent executors are archived for the frozen application records. Imported equations and sections resolve through its semantic labels. Historical v94 and v40.03r24 records are comparison inputs, with their original statuses preserved; they do not replace this foundation.

Literature note. Proof discipline and released foundation: Poyau [13, 14, 15].

2 Imported Relations and Conditional Support Arithmetic

2.1 Distinction, motif and instance evidence

Setup. Main distinguishes Null sameness from committed finite events (Main (1), Main (3)). The branch/hinge/counterbranch motif is

$$M_{\varepsilon,m} = \varepsilon\, 0_H (0_D)^m (-\varepsilon),$$

with mandatory hinge and optional dwell (Main (5)). A faithful atomic edge changes one bit of Q_4 (Main (8)). These relations provide the motif template used by the visual projection.

Derivation. For a native instance, a declaration must bind its scope, lineage and construction:

$$\mathbf{a} \vdash (F, R, \partial_T R, \omega), \qquad B = (\partial_T F, \omega). \tag{1}$$

A certified closure additionally binds the retained-start return witness and at least two committed phase-bearing traversals (Main (13), Main (15)). The admitted motif record in Main (33) and traced admission in Main (37) are prerequisites, not consequences of writing an elemental label.

Specialization. This reference supplies grouping, local tuples and conditional expressions for labelled rows. It does not supply the scope-bound instance evidence in (1). Names identify the inherited distinctions or the declared reference records; they introduce no additional object or proof condition. Conversely, names cannot discharge the hypotheses of the construction they identify.

Falsification. A formatted support declaration represented as a successful certificate, or a plotted line represented as an executed edge without its trace, exceeds this profile.

Provenance. Main (5), Main (13), Main (33), Main §5.17.10.

2.2 Support polynomial and the reflection-duration domain

Setup. A support expression $\Sigma = (s_1, \ldots, s_k; L)$, with clauses $s_i = p_i{:}q_i$, is formal syntax (Main (36)). The declared finite support model is

$$\Pi(p{:}q) = q^p + q, \qquad p, q \in \mathbb{Z}_{\geq 1}. \tag{2}$$

Its provenance is Main (35) and Main (177). Admission of the proposed form remains

$$K_{\text{prop}} \xrightarrow{HT(K;B)} K_{\text{adm}}(B). \tag{3}$$

Derivation. For an admitted instance in the domain of the inherited accessor,

$$RD_\gamma(B) = \Pi^-_\gamma(B) + H_{\mu,\gamma}(B) + \Pi^+_\gamma(B). \tag{4}$$

Under equal branch supports and $H_{\mu,\gamma} = 1$,

$$RD^{(1)}_\gamma(B) = 2\Pi(p{:}q) + 1. \tag{5}$$

These are Main (31), Main (32) and Main (179). Evaluating the right-hand expression numerically does not establish its accessor prerequisites.

Specialization. Table 1 preserves the exact polynomial values. Its last column is the conditional single-hinge expression; it is not evidence that the corresponding elemental instances have been admitted.

Table 1: **Imported support polynomial and conditional single-hinge expression.**

Clause	q^p	$\Pi = q^p + q$	$2\Pi + 1$
1:2	2	4	9
3:1	1	2	5
3:2	8	10	21
3:3	27	30	61
3:4	64	68	137

The ordered count basis is

$$\mathcal{B}_{\text{curl}} = (3{:}3,\ 3{:}4,\ 1{:}2), \tag{6}$$

and its local declaration expands as

$$a{:}b{:}c :: L \equiv (3{:}3)^a :: (3{:}4)^b :: (1{:}2)^c :: L. \tag{7}$$

These local counts are distinct from the nuclear counts $(Z, A-Z)$ in §4. Main's type distinctions remain unchanged: Dimonon = 2 Monon ≠ Dion, and tetrion, tetron, tetratrion and tetrad denote different constructions.

Falsification. The single-hinge expression cannot replace (4) for unequal branches or a non-unit hinge. A declaration string cannot supply a missing construction witness or cycle-valued compression.

Provenance. Main App. B.4.1, Main App. B.1.10, Main (34), Main (40), Main (43).

3 Enclosure Membership and the Visual Model

3.1 Even enclosure sequence and cumulative boundaries

Setup. The inherited enclosure grammar distinguishes E_2 (duon), its open and seated states $E_2^0, E_2^{\pm}$, the inner E_4 tetron, and outer even support enclosures C_{2m}, $m \geq 3$. The present chart uses C_6 through C_{22} as nested enclosure interfaces. Their displayed geometry represents declared enclosure membership; no measured radius or metric scale is assigned. The semantic sources are Main §5.7 and Main App. B.3.

Derivation. The coord-0 declaration and filled boundary form are

$$0 \mid - :: Z : A : 0 : L\langle a : b : c :: L\rangle, \tag{8}$$

$$0 \mid - :: Z : A : 0 : 2m\langle m : m : 0 :: 2m\rangle. \tag{9}$$

Following Main §5.7.13,

$$\Gamma_m = \sum_{k=3}^{m} 2k = m(m+1) - 6, \qquad \Gamma_2 = 0, \tag{10}$$

and

$$\Gamma_{m-1} < Z \leq \Gamma_m, \qquad s = Z - \Gamma_{m-1} \tag{11}$$

select the unique group and seat. Here $Z = 1, \ldots, 118$ identifies the labelled elements; integers 119–126 are unfilled reference-slot indices in the same grouping.

The local tuple is

$$a(s) = \lceil s/2 \rceil, \qquad b(s) = \lfloor s/2 \rfloor, \qquad c(s) = s \bmod 2. \tag{12}$$

Thus $c = 1$ records odd local parity and $c = 0$ even local parity. It does not determine an assigned acceptor, neutral or shedding class. For example, Li has $s = 3, c = 1$ but an assigned neutral class; O has $s = 2, c = 0$ but an assigned acceptor class. The tuple and those input classifications are distinct relations.

Specialization. The nine groups in Table 2 contain 126 reference slots, of which 118 carry element labels:

$$118/126 = 59/63, \qquad N_{\text{labelled}}(\mathsf{C}_{22})/22 = 14/22 = 7/11.$$

These are labelled-reference occupancy fractions. No retention duration or physical limit beyond 118 follows from them.

Table 2: **Declared enclosure groups and local role positions.** The last endpoint is an unfilled reference slot.

m	$2m$	Γ_{m-1}	Γ_m	Index range	Hinge $s = m$	Boundary $s = 2m$
3	6	0	6	1–6	3 Li	6 C
4	8	6	14	7–14	10 Ne	14 Si
5	10	14	24	15–24	19 K	24 Cr
6	12	24	36	25–36	30 Zn	36 Kr
7	14	36	50	37–50	43 Tc	50 Sn
8	16	50	66	51–66	58 Ce	66 Dy
9	18	66	84	67–84	75 Re	84 Po
10	20	84	104	85–104	94 Pu	104 Rf
11	22	104	126	105–126	115 Mc	126 –

3.2 Relations before layout

Setup. The visual model preserves $Z \mapsto (m, s, a, b, c)$. In each enclosure, $1 \leq s < m$ projects the first branch, $s = m$ the hinge, $m < s < 2m$ the counterbranch, and $s = 2m$ the filled-tuple boundary. This projection does not execute inter-element edges.

Derivation. Nested outlines express enclosure membership; labels express local order and role. A spiral can project seat order while preserving membership, but its curvature supplies neither a temporal trajectory nor a transition witness. Assigned classes retain their input meaning from §4.

Falsification. Changing preserved Z, m, s, equating odd parity with acceptor status, or treating unfilled slots as physical non-retention contradicts this model. Field count $n_F \log n_F$ in Main (91) has a separate scope; ring counts cannot supply its field-family witness.

Provenance. Main §5.7.10, Main §5.7.13, Main §5.8, Main §5.7.12, (11), (12).

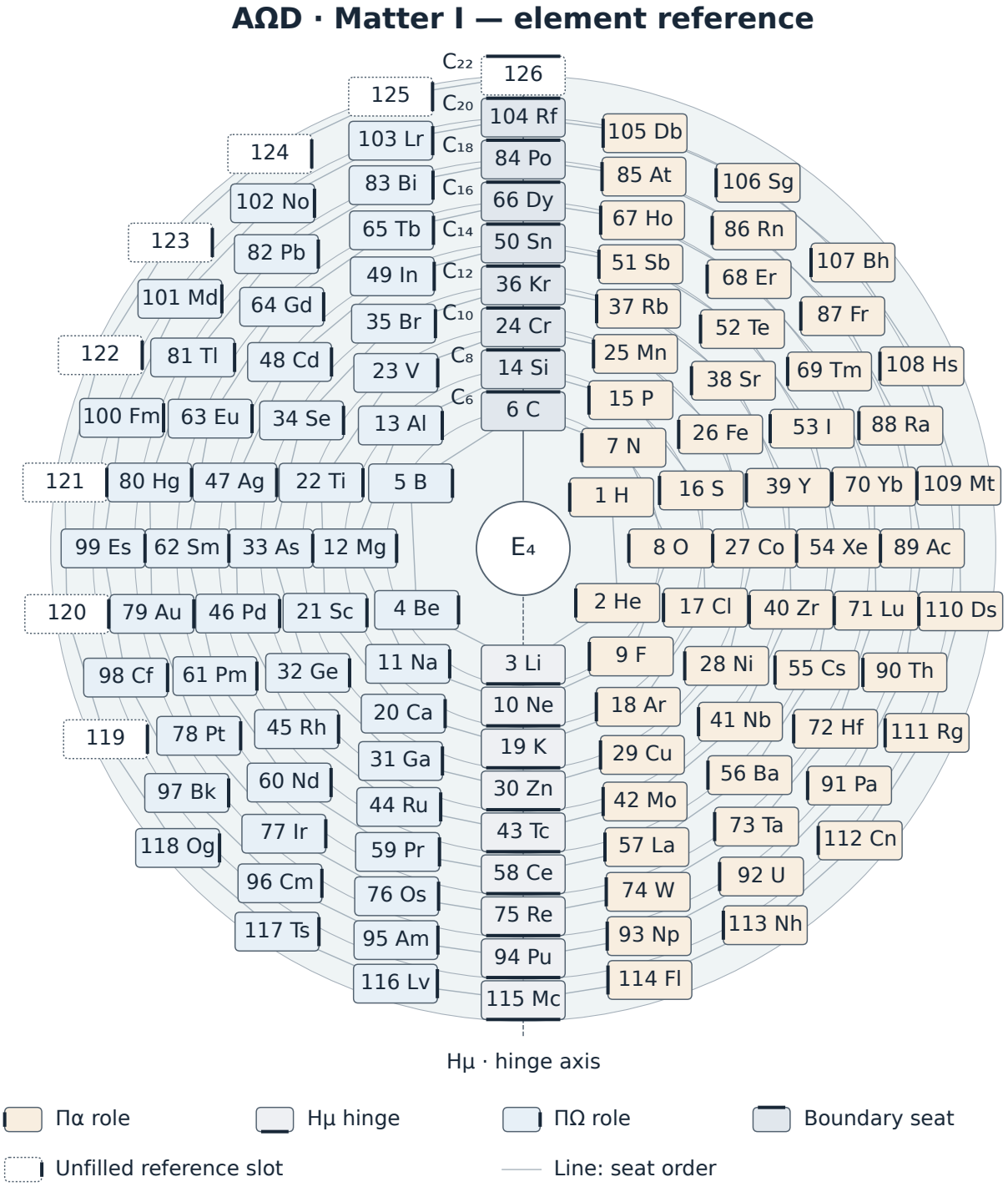


Figure 1: **Matter I static relational reference.** Nested enclosure membership and repeated branch/hinge/counterbranch roles organize all 118 element labels. The line is a seat-order guide. Positions 119–126 are unfilled reference slots in the declared grouping. Geometry supplies neither physical scale nor an executed network.

4 Conditional Support Expressions and Assigned Inputs

4.1 Inherited fusion domain and the present calculation

Setup. For admitted forms F, G with certificate-bound cycle-valued compressions $\widehat{F}, \widehat{G}$, Main defines

$$\mathrm{Fuse}_B(F, G) = \mathrm{Close}_B(\widehat{F}\,\mu\,\widehat{G}) \tag{13}$$

and

$$RD_{\mathrm{fuse}}(F, G; B) = \Pi^{-}_{\widehat{F}}(B) + H_{\mu,\widehat{F},\widehat{G}}(B) + \Pi^{+}_{\widehat{G}}(B). \tag{14}$$

Their exact predecessors are Main (40), Main (115) and Main (116). These operators retain their domains. The present elemental table supplies no inner or outer fusion certificate and does not assert an instance of (13).

Derivation. The declared reference operands use the exact polynomial values in Table 1:

$$u_{-} = 30Z + 68(A - Z), \qquad h = 1, \qquad u_{+} = 4s. \tag{15}$$

Define their conditional support sum

$$U(Z, A, s) = u_{-} + h + u_{+} = 30Z + 68(A - Z) + 1 + 4s. \tag{16}$$

This preserves the scalar expression formerly stored as `rd_fuse`. The active field `support_sum` names its established arithmetic scope. Equality to a native reflection-duration accessor additionally requires the inherited instance prerequisites.

Specialization. The nuclear count pair $(Z, A - Z)$ in (15) is distinct from the local tuple (a, b, c) in (12). At fixed Z, changing the supplied integer A leaves m, s, a, b, c unchanged and gives

$$U(Z, A + \Delta A, s) - U(Z, A, s) = 68\Delta A.$$

For carbon, $Z = s = 6$, the conditional choices $A = 12, 13, 14$ give $U = 613,681,749$. This is an algebraic illustration, not a sourced isotope catalog, stability statement or decay/fusion transition.

The reference A values are carried from the reviewed candidate's input list without alteration. Their original per-element selection policy was not recorded. The input provenance identifies that candidate source and this limitation; no broad AME or NIST citation establishes each selected A. Reference A is an integer nucleon-count input, not standard atomic weight. Projected A values for slots 119–126 remain explicitly illustrative fixture operands, not observed nuclide records. The inputs are recorded in `manual2/data/m1/reference_inputs.csv`, `model_inputs.json` and `input_provenance.json` in the same directory.

Falsification. A change in A that changes the grouping or local tuple violates this specialization. Calling a numerically correct U a certified fusion duration without its input and outer certificates exceeds its domain.

Provenance. Main (35), Main (116), (15), (12).

4.2 Assigned classifications, residuals and occupancy

Setup. The fixture keeps the candidate's assignment sets:

$$\mathcal{A} = \{1, 8, 9, 16, 17, 34, 35, 52, 53\}, \qquad \mathcal{X} = \{43, 61\} \cup \{84, \ldots, 118\}.$$

They define assigned acceptor and assigned shedding classes. They are input classifications; they are not derived from Z, A, parity, or an executed native calculation. Let $L = 2m$, $C^{\mathrm{fix}}_{\mathrm{close}} = 3L$, and $w^{\mathrm{fix}} = L$.

Derivation. With assigned pressure P^{fix}, retain the residual arithmetic

$$S^{\mathrm{fix}} = P^{\mathrm{fix}} - C^{\mathrm{fix}}_{\mathrm{close}}, \qquad X^{\mathrm{fix}} = \max(0, S^{\mathrm{fix}}). \tag{17}$$

The pressure input is $C^{\mathrm{fix}}_{\mathrm{close}} - 3$ on $\mathcal{A}$, equal to $C^{\mathrm{fix}}_{\mathrm{close}}$ on the remaining labelled neutral rows, and $C^{\mathrm{fix}}_{\mathrm{close}} + 3u_Z$ on $\mathcal{X}$, where

$$u_Z = 1 \ \ (Z = 43, 61), \qquad u_Z = \lfloor (Z - 84)/6 \rfloor + 1 \ \ (84 \le Z \le 118).$$

For illustrative unfilled slots it is $C^{\mathrm{fix}}_{\mathrm{close}} + 3(Z - 118)$. Assigned duration values are respectively L, $L+2$, $L-1$, and 0. Assigned route and Yro labels are carried as fixture data. These values do not demonstrate retention, instability, transfer or non-retention.

The arithmetic uses the form of Main (108) and Main (109); it does not supply their native pressure, field and window evidence. Likewise, a declared route label does not execute the sheddic partition, lift or higher-scope fusion in Main (111), Main (120) and Main (122).

Specialization. The 118 labelled rows have disjoint assigned residual classes: 72 neutral, 9 deficit/acceptor and 37 positive-surplus/shedding. Boundary position is a separate attribute: eight labelled endpoints are $\{6, 14, 24, 36, 50, 66, 84, 104\}$. Po and Rf are both endpoints and assigned shedding rows. Adding endpoint counts to route-class counts would double-count those attributes. Of the 72 neutral rows, six are endpoints and 66 are not.

The C_{22} fraction 7/11 records 14 labelled slots out of 22. Slots 119–126 are unfilled in this reference; their assigned fixture values establish neither a physical cutoff nor an impossibility of further elements.

Falsification. The assignment table and its equations must agree exactly. Positive X^{fix} for negative S^{fix}, or a changed assigned set without a new input binding, fails the fixture. Agreement with these definitions is not empirical validation.

Provenance. Main (108), Main (109), Main §5.7.12, Main §5.6.6.

4.3 Complete reference ledger

The authoritative generated ledger is `manual2/data/m1/matter1_elements_1_118_ledger.csv`. It binds each static declaration to its input record, distinguishes labelled elements from unfilled slots, and carries pending admission and unexecuted native status. The support declaration is syntax; the support sum and assigned fields retain the scopes above. Assigned route codes are `neutral`, `deficit`, `surplus` and `unfilled`; group-boundary and named-anchor flags are independent attributes. The generated table and figure consume the same validated data. Declared compatibility copies must compare equal.

The complete 118-row reference ledger is retained in App. D; the compact chart remains the primary overview.

5 Verification Scope and Pending Native Evidence

5.1 Executed static checks and inherited audit obligations

Setup. The static checker verifies stored inputs and generated artifacts without rewriting them. Generation is a separate explicit operation. Exact grouping, local tuples, support sums, assignment residuals, identities and data/figure agreement are testable here. Their success does not certify the six native hooks in Main §5.17.

Derivation. Table 3 separates the available static checks from the instance evidence required for native claims. A check result is reported only after that check executes; this table describes scope, not an unrecorded run count.

Table 3: **Verification scope.** Element-wide native obligations remain pending.

Inherited hook	Static reference check	Native evidence not supplied
Spine	Declared role mapping and operand sum	Retained occurrences at each finite depth
Seat/enclosure	Nine boundaries, tuples and labelled occupancy	Scoped field-family and routing witness
Sheddic path	Assigned class, route and residual consistency	Realized path and retention window
Cycle	Support syntax and pending status	Construction, retained start and return certificate
Fusion	Conditional operand and residual equations	Admitted inner forms and outer closure certificate
Naming	Stable declarations, record kinds and input bindings	No name substitutes for instance evidence

Specialization. A future native D.E.C. record has the inherited form

$$(B,\ v,\ e,\ v_e,\ \sigma(e),\ \mathrm{adm}(e;B),\ w(e;B),\ P(e;B),\ \mathrm{route}(e;B)).$$

The present element reference ledger does not instantiate it. The separately declared finite model in §6 executes its own documented predicates and histories; it is not an element-wide D.E.C. ledger. Native execution would require bound Q_4 Hamming-1 traversals, scope and window, admissibility and exact weights, normalized kernels on nonempty positive-weight support, and reversed-edge SADAR pairs:

$$\sum_e P(e;B) = 1, \qquad \mathrm{adm}(e;B) = 0 \Rightarrow P(e;B) = 0, \qquad \mathfrak{S}_B(C_{\bar{e}}) = -\mathfrak{S}_B(C_e).$$

An empty admissible support does not define a normalized kernel. Each retained-return witness must bind its own retained identity, scope and lineage. Fold and comparison consumers must additionally preserve explicit compatible maps between their distinct scopes and lineages. None of these obligations is discharged by this reference's assignment tables.

Falsification. Actual negative probes must reject changed values, duplicate or missing identities, mismatched mirrors, invalid status promotion and altered provenance without replacing the input bytes. In particular, a forged certificate-like string must not turn pending admission into a pass. Unknown command flags must fail, and the verification command must not regenerate corrupted input.

Publication checks bind Main source/PDF/AUX/FLS before resolving semantic references. Figure checks cover all 126 marks at final size, including the eight unfilled slots. These are distinct from native dynamics or empirical tests.

Provenance. Main §5.17.1, Main §5.17.4, Main §5.17.5, Main §5.17.7, Main §5.17.8, Main §5.17.10, Main (66), Main (96).

Literature note. D.E.C. row grammar and execution protocol: released AOD Manual v41.02, §4 [15].

6 Folded Construction and Relational Time

6.1 Distinction, handed hinge and retained lineage

Setup. The schematic sameness presentation $\cdots xxx \cdots = x$ belongs to Main §1.1. Its occurrences are not previously distinguished events or ticks. The first distinction and its anchor precede the finite construction below. On an oriented construction, the shorthand $x(x)x$ resolves as $x^{-}(x^{0})x^{+}$: branch, retained hinge and counterbranch. Reflection exchanges the sides and preserves the hinge. This Manual instantiates those relations; it does not replace Main's ontology with a runtime string-reduction rule.

Derivation. Handedness records branch order through the hinge. The support accessor inherited from Main (31) is

$$RD_B(C) = \Pi_C^- + H_{\mu,C} + \Pi_C^+.$$

Equal scalar support does not identify oppositely oriented constructions. A duration realization, a count of traversals and a count of retained returns require their respective maps. In particular, a structural hinge is mandatory even when optional hinge dwell is zero; it need not be a seventh Hamming-1 traversal.

Folding relates distinct inner and outer scopes through explicit maps

$$f_- : (B_-, i_-, \Gamma_-) \longrightarrow (B_o, i_o), \qquad f_+ : (B_+, i_+, \Gamma_+) \longrightarrow (B_o, i_o).$$

These maps preserve the source identity, scope, lineage, orientation, history and certificate. A proposed outer boundary requires its own retained start and return witness Γ_o. Exposure follows the maps back to the inner records. Neither common endpoint coordinates nor a handedness flip supplies an outer certificate. Cross-scale scope equality is not required.

Provenance. The source obligations are Main (5), Main (6), Main (46), Main (41) and Main (42).

6.2 An injective two-face example

Setup. Declare six distinct vertices of Q_4, with each committed base traversal changing one coordinate:

vertex	A	B	C	D	E	F
address	0000	1000	1100	0100	0010	1010

Take the oriented square faces $C_- = (A, B, C, D, A)$ and $C_+ = (B, A, E, F, B)$. They share only the coordinate edge AB, with opposite orientation. Assign interface roles to A, B, negative-side roles to C, D, and positive-side roles to E, F. The role projection is separate from the coordinate path.

Derivation. Cancellation in the oriented boundary chain gives

$$\partial(C_- + C_+) = (A, E, F, B, C, D, A).$$

The shared interface remains present in the incidence and history records. Cancellation of its outer coefficient does not erase that record. The two legs (A, E, F, B) and (B, C, D, A) each contain three Hamming-1 traversals. For n declared repetitions, let x denote the number of committed traversals in one leg family. Then

$$x = 3n, \qquad 2x = 6n. \tag{18}$$

Here n counts repetitions of this specified pattern; one counted tick is one committed Hamming-1 traversal. The factor three follows from the enumerated edges, not from naming three roles. The persisted illustrative ledger contains three repetitions, nine traversals per leg and eighteen distinct event identities. It asserts no native closure or beat count.

Specialization. Main's two-nonempty-leg bound in Main (28) constrains an already admitted closure. It neither selects this six-edge construction nor fixes a universal duration. A four-edge square is also a closed Hamming-1 walk. Conversely, closure of a geometric walk alone does not admit a native motif. One accepted retained return contributes one closure phase; later distinct returns of that same lineage supply beats as specified by Main (21). Replaying a record supplies neither a new traversal nor a new return.

If an additional model identifies each branch support operand with $3n$, its RD expression is $6n + H_\mu$, with $6n+1$ only in the single-hinge specialization. That support statement is distinct from the six-edge traversal count. The mathematical constant π, physical seconds and an atomic frequency do not enter this finite counting model.

Falsification. The checker rejects the coordinate aliases $E = D$, $F = C$ before counting: they make the face boundaries coincide with opposite orientation rather than leave the claimed six-edge boundary. It also rejects a reversed interface, a non-Hamming-1 edge, a missing role, duplicate external event identities, reserialized/recounted events, wrong scope or lineage, and unsupported status promotion. Checks run against persisted inputs without changing their bytes. The generator is a separate command.

Provenance. `manual2/data/m1/folding/two_square_fold.json`, `event_ledger.json` and `folding_report.json`; Main (8), Main (74) and Main (15).

6.3 Declared recurrence and mapped comparison

Setup. The separate finite model commits the next oriented edge at each update cut. A distinct start sample follows a completed return; optional hinge dwell contributes a zero-edge sample. These update cuts are the declared model's ordering domain, not seconds or a newly derived universal clock. Two inner square lineages, each with four retained returns, are embedded into each of two oppositely oriented outer lineages. The outer traces follow the six-edge boundary with their own starts and returns. Their inner maps remain explicit.

The model declares zero optional dwell for the positive outer orientation and one sample for the negative orientation. It declares phase biases $b_+ = 1/4$, $b_- = -1/4$, four comparison contacts, and three reporting cuts. These are fixture inputs. No target frequency, desired physical period or stochastic kernel is read by the generator.

Derivation. The inner completion cuts are $4, 9, 14, 19$. The positive outer completion cuts are $26, 33, 40, 47, 54$; the negative outer cuts are $27, 35, 43, 51, 59$. Thus both outer traces have six Hamming-1 traversals and the same declared support sum $6 + 1 = 7$, while their completion intervals are respectively seven and eight update cuts. Support and recurrence intervals retain different types even where their numbers coincide.

Each return witness binds its own start, hinge, events, scope and lineage. Event and return identities are unique; the first retained return supplies the origin, and subsequent returns supply beats. Two identity-preserving maps bind committed return prefixes to the comparator scope. Repeated contacts are required. For the windows in Table 4, define

$$\mathrm{biz}_{+|-} = \frac{\Delta \,\mathrm{bip}_+}{\Delta \,\mathrm{bip}_-}, \qquad \psi = b_+ - b_- = \frac{1}{2}, \qquad \Lambda(v) = N_+(v) - N_-(v) + \psi.$$

Here $N_\pm(v)$ are cumulative reclosure counts from each lineage's origin; they equal the window beat increments for these windows starting before both outer origins. A later window does not reset unwrapped lag. The reference counts are positive. Exchanging the comparator direction gives reciprocal cadence and negated bias/lag. Equal cadence in the first window does not remove its nonzero bias or lag.

Table 4: **Executed declared-model comparisons.** Drift is the change from the preceding reported unwrapped lag; its first value is undefined.

Window	Beat counts $(+, -)$	Cadence ratio	Unwrapped lag	Drift
$(20, 43]$	$(2, 2)$	1	$1/2$	undefined
$(20, 47]$	$(3, 2)$	$3/2$	$3/2$	1
$(20, 54]$	$(4, 3)$	$4/3$	$3/2$	0

Specialization. The declared contact support is two traversals plus one structural hinge, giving three support units. An explicit additive contact-duration map adds reflection counts $(1, 1)$ in one direction and $(2, 2)$ in the other. Its directional duration counts are five and seven. These are contact durations under a declared map, not the outer support accessor or empirically measured periods. The model executes finite Hamming-1, hinge, own-return and fold-map predicates, while complete motif admission and native atomic execution remain pending.

Provenance. `temporal_model.json` declares the update and comparison inputs; `temporal_trace.json` stores events, witnesses, maps and contacts. Source anchors: Main (54), Main (55), Main (58), Main (61) and Main (63).

6.4 Causal returned-current response and exposure

Setup. For the same committed comparison histories, declare boundary weight $C_3 = 1$, oriented current $i = 1$ and forward duration participation $\rho_\omega^D = 5$. Use the preceding signed lag in the inherited returned-current expression. The three compatible capacities are declared inputs $10, 10, 3$, not predicted material properties. Responses commit strictly after the reporting cut that supplied their inputs.

Derivation. This specialization evaluates

$$\mathfrak{S} = C_3 \rho_\omega^D \Lambda i, \qquad P^{\mathfrak{S}} = C_3 \rho_\omega^D |\Lambda|, \qquad X = \max(0, P^{\mathfrak{S}} - C^{\mathrm{close}}).$$

It yields $\mathfrak{S} = 5/2, 15/2, 15/2$, residuals $-15/2, -5/2, 9/2$, and excesses $0, 0, 9/2$. The first two windows have nonzero lag and no positive excess. A separately bound reversal pair preserves boundary, window, history and duration participation while reversing edge orientation and signed current; its returned-current values are antisymmetric. This pairing is distinct from exchanging the two directional duration maps.

The responses commit at cuts $44, 48, 55$. The last loses positive-outer retention under the declared capacity rule, records an excess route and exposes the preserved inner maps and certificates. This is a model bridge between a typed excess and loss of outer retention; it is not an identity valid for every boundary. Routed-material reclosure and any radiation identification remain pending.

Falsification. Persisted tests reject forged finite certificates, missing structural hinge, omitted/substituted inner maps, recycled inner returns, replayed or repackaged returns, one-contact comparisons, forced duration symmetry, same-step self-causation, mismatched reversed-edge witnesses and unsupported status changes. The finite model's exact reproduction is tested independently of native admission. A source-derived dynamics rule, complete motif/accessor admission and an atomic observation map are still required before these fixture cadences or capacities can become physical predictions.

Provenance. `temporal_report.json` separates declared execution, finite witness scope, native admission and empirical status. Source anchors: Main (94), Main (96), Main (101), Main (108) and Main (109).

Literature note. The inherited distinction, fold and recurrence definitions are in the released AOD foundation [15]. The finite fixture and update model in this section are Manual applications.

7 Staged Hyperfine Application

7.1 Atomic identity and sourced recurrence

Setup. The proposed application compares recurrent atomic and oscillator systems through one common fusion/reclosure model. Element placement alone does not identify a hyperfine state: hold Z, isotope A, charge, electronic state and nuclear spin fixed, and distinguish the coupled states. The present inventory has two neutral isotopes, four states and two transitions. It adds no placement to the 118-element reference and changes no reference-A input.

Table 5: **External atomic inventory.** Frequencies are source records, not predictions of the finite folding model. Both clock transitions have $m_F = 0$ at each endpoint.

Nuclide	Electronic state; I, J	Transition	Frequency record
^{1}H, $Z = 1$	$1s\,^2S_{1/2}$; $1/2, 1/2$	$F = 1 \leftrightarrow 0$	1 420 405 751.768 Hz; reported one-sigma uncertainty 0.002 Hz; historical 1970 measurement
^{133}Cs, $Z = 55$	[Xe]$6s\,^2S_{1/2}$; $7/2, 1/2$	$F = 4 \leftrightarrow 3$	9 192 631 770 Hz; exact defining value; realization uncertainty is a separate quantity

Derivation. Hellwig et al.'s printed page 200 reports the hydrogen value with uncertainty 2×10^{-3} Hz. The exponent is bound to the retained page image, correcting the earlier plan's 0.02 Hz transcription. This is a dated measurement, not a current recommended value. Dividing the quoted decimal by the

defining caesium value gives the external reference ratio

$$r_{\mathrm{H/Cs}}^{\mathrm{ref}} = 0.1545156803086000, \qquad u_r^{\mathrm{reported}} \simeq 2.1757 \times 10^{-13}.$$

The implementation retains exact rational arithmetic of the transcribed decimals; the measured hydrogen frequency is not physically exact. The uncertainty shown divides the paper's reported one-sigma uncertainty by the exact defining denominator. It is not a new uncertainty evaluation or an independent recombination of the two hydrogen evaluations.

The inventory retains the common NBS-III caesium reference and the paper's wall, cavity-pulling, spin-exchange, magnetic-field and second-order Doppler correction context. Quantitative covariance and separate realization uncertainties remain unresolved where not supplied. A missing quantity is recorded as unknown, not zero. A new precision comparison must supply the applicable conditions and shared-reference uncertainty budget.

Specialization. Atomic interrogation prepares coherence and compares its later phase with an oscillator; population measurements provide feedback to that oscillator. The counted recurrence is the stabilized oscillator cycle. A detection count, Rabi oscillation, spontaneous waiting time and hyperfine frequency are different observables. Relating oscillator cycles and quantum phase to certified native reclosures requires a physical observation map.

Provenance. `manual2/data/m1/hyperfine/target_inventory.json` binds nuclide, state, transition, source and uncertainty records. `sources/hellwig1970-page200.png` preserves the transcription evidence. The report retains the ratio's measured-input scope.

Literature note. Historical measurement and state labels: Hellwig et al. [3]. Atomic data: NIST [10, 16]. Defining reference: BIPM [1]. Clock operation: NIST [8].

7.2 A common model and its remaining observation map

Setup. The application hypothesis is a reconfiguration between two states of the same isotope, bound to one frozen fusion/reclosure model. Proposed before/after configurations must preserve the isotope identity and the actual retained inner lineages. A transition need not increase element number, mass number or folding depth. A detected transfer can be identified as native shedding only after its source, destination, direction, window and measured observable are mapped.

Derivation. The atomic inventory (P4a), applicable construction (P2) and coupled recurrence (P3) are all prerequisites of the physical observation map (P4):

$$(\mathrm{P4a}, \mathrm{P2}, \mathrm{P3}) \longrightarrow \mathrm{P4}.$$

The finite declared-model checks of §6 instantiate selected construction and comparator predicates. They do not supply isotope-specific native admission, a physical state/configuration map, or an independently derived update rule. Thus P4 remains pending. Chirality or a ternary role cannot be equated with m_F by visual resemblance, and a separate fitted law for each spectral line cannot count as a prediction of one common model.

The next native AOD physical comparison requires admitted atomic configurations, their recurrent histories and coupling records, a map from measured phase/cycles to native phase/returns, a metrological duration or frequency map, and its calibration and uncertainty model. Until those premises are established, the native predicted frequency ratio and empirical score remain absent. Separately declared

conditional models and conventional benchmarks can execute under their own identified premises, as in §8. The same distinction applies to a proposed occurrence of the mathematical constant π: no physical result is inferred from its use in a drawing or angular convention.

Specialization. The hydrogen target was exposed during the discussion and supplied plans before this implementation. `target_exposure.json` records exposure by 26 October 2026; an earlier exact first-exposure time is unknown. The known pilot is retrospective. A subsequent model freeze does not establish independent selection or blindness. A calibrant cannot also be scored as a held-out target. Independent prediction requires a separately justified target and selection history. Frequency-only data do not test phase bias, lag or drift.

Falsification. Persisted hostile tests reject changed atomic foreign keys, the incorrect hydrogen uncertainty, omitted correction lists and covariance statuses, unsupported native or empirical promotion, a policy allowing target access during candidate selection, a calibrant labelled held out, detection/cycle substitution, and per-line fitted models advertised as independent predictions. These are actual schema and provenance guard tests. Their success is not empirical agreement. Every consumed status is checked; missing statuses fail rather than silently becoming zero or success.

Provenance. `observation_map.json` records the P4a/P2/P3 dependencies and pending physical map. `hyperfine_report.json` lists unresolved dependencies and null prediction/score fields. Native recurrence, comparator and duration definitions are inherited from Main (61), Main (63) and Main (57).

Literature note. External atomic observations remain downstream of the released AOD definitions [15]; the historical target remains bound to Hellwig et al. [3].

8 State-Dependent Evolution and the SI Prediction Experiment

8.1 A response that changes later events

Setup. The second finite model retains the two-face geometry of §6. Its state contains the oriented boundary position, structural-hinge sample, retained outer identity, committed return and contact records, pending response and inner scope maps. The declared update consumes that state at the next integer cut. It instantiates the form of Main (54); the particular sequencer and capacity rule are application inputs.

Derivation. At a comparison cut, the engine calculates a burden from the committed histories and records a response for the following cut. A retention loss then removes the affected outer identity from subsequent traversal and contact eligibility. An unfinished traversal is not promoted to a return. Previously completed returns and the inner maps remain in the trace. Re-admission would require a separate construction; this model supplies none.

The controlled pair changes only the declared capacity. Its common prefix is identical before the response, while later return and contact records differ. This establishes causal state dependence within the finite model. The frozen first version remains available as the original response-calculation fixture; its precomputed histories are unchanged.

In the persisted pair, capacity changes from 100 to 2 at cut 40, and loss commits at cut 41. The affected returns at cuts 47, 54, 61 and 68 disappear; completed contacts decrease from eight to two. With the

first completion retained as the lineage anchor, the finite-window reclosure-count ratio changes from 6/5 to 2/5. These are counts over the declared window, not a stationary frequency estimate.

Specialization. Integer cuts, six-edge boundary traversals, hinge support, completed returns and cadence remain distinct types. The finite comparison uses Main (61) with its explicit histories and windows. It does not identify an outer recurrence with either an atomic transition or one SI second.

Falsification. The verifier rejects a post-loss affected return, an unavailable contact, altered earlier history, missing inner map, forged status or stale source binding. A changed future response cannot modify an earlier committed prefix. These checks operate on persisted counterexamples and do not repair their inputs.

Provenance. `manual2/data/m1/stateful/`; the separate model and trace verifier preserve `manual2/data/m1/folding/`. Source ownership follows Main (52), Main (53) and Main (54).

8.2 The atomic observable is an additional construction

Setup. The designated experiment uses Cs-133 as the frequency-scale reference and H-1 as the comparison target. Both configurations must belong to one declared atomic dynamics and observation law. The law must calculate a dimensionless ratio $\widehat{R}_{\mathrm{AOD}}$, before the hydrogen evaluation value is read. The external state inventory of §7 identifies the transitions; it does not supply that law.

Derivation. Once the physical observation map is established, ideal-reference reporting gives

$$\widehat{\nu}_{\mathrm{H}} = \widehat{R}_{\mathrm{AOD}}\,\Delta\nu_{\mathrm{Cs}}, \qquad \Delta\nu_{\mathrm{Cs}} = 9\,192\,631\,770\ \mathrm{Hz}, \qquad \widehat{T}_{\mathrm{H}} = \widehat{\nu}_{\mathrm{H}}^{-1}. \tag{19}$$

The Cs definition fixes the unit scale. It does not determine $\widehat{R}_{\mathrm{AOD}}$. Likewise, a ratio of finite return counts becomes a physical frequency ratio only through the declared atomic observation map.

Specialization. The cited source premises and the present atomic contract do not supply an instantiated Cs/H configuration-to-observable law. The executable underdetermination experiment therefore keeps its finite histories and Cs anchor fixed while varying the unconstrained observation relation. Different resulting hydrogen reports demonstrate that these premises do not select a unique atomic prediction. The alternatives are diagnostic countermodels, not proposed measured frequencies or fitted atomic laws.

Specifically, let r be the finite count ratio and assign the two lineages hypothetical labels $Z = 1, 55$. The unselected observation family

$$g_\lambda(Z) = 1 + \lambda\frac{55 - Z}{54}, \qquad \lambda > -1, \qquad R_\lambda = r\,\frac{g_\lambda(1)}{g_\lambda(55)} = r(1+\lambda) \tag{20}$$

preserves the Cs normalization for every member while allowing every positive ratio. The witnesses $\lambda = 0, 1$ give $6/5, 12/5$ for the baseline and $2/5, 4/5$ after intervention. This proves underdetermination by the implemented finite constraints, not a general impossibility of an atomic AOD construction.

The minimum additional premise is a justified common atomic construction and observable law that fixes the relative Cs/H response without importing the tested hydrogen splitting, its reciprocal period or an equivalent gain. Selecting one countermodel by agreement with the known target would fit that target. The atomic prediction objective remains incomplete until this premise is supplied and tested.

Calibration. For the conventional benchmark, dimensionless spin operators obey

$$\frac{H}{h} = \frac{A_{\mathrm{hfs}}}{h}\, \boldsymbol{I}\cdot\boldsymbol{J} + \frac{\mu_B}{h} B(g_J J_z + g_I I_z), \qquad \nu(B) = \sqrt{\Delta^2 + [(g_J - g_I)\mu_B B/h]^2} \tag{21}$$

in the $J = 1/2, m_F = 0$ block, with $A_{\mathrm{hfs}}/h = \Delta/(I + 1/2)$. The imported zero-field splittings are $\Delta_{\mathrm{H}} = 1\,420\,405\,751.768\,\mathrm{Hz}$ and $\Delta_{\mathrm{Cs}} = 9\,192\,631\,770\,\mathrm{Hz}$. The declared approximation uses $g_J = 2, g_I = 0$, omitting nuclear and bound-state corrections. At the stipulated field $B = 10^{-4}\,\mathrm{T}$, the computed shifts are approximately 2758.292 Hz for H and 426.200 Hz for Cs. These are conventional numerical benchmark outputs, with incomplete field and approximation uncertainty.

The separately labelled conventional hyperfine calculation uses this spin Hamiltonian and imported transition constants to check numerical machinery. Reproducing an imported splitting verifies that calculation; it does not predict the splitting from AOD. A conditional ideal-reference result and an actual clock realization have different uncertainty inputs. The reference definition carries no uncertainty; an employed device, transfer or calibrated observation map does.

Falsification. The prediction stage binds model, histories, observation assumptions and calibration hashes before the comparison stage reads the target. Changing an evaluation value must leave the prediction artifact unchanged and alter only the corresponding residual. A controlled finite-dynamics intervention must affect the diagnostic mapped observable at a later cut. These software properties do not supply the missing atomic law. The known hydrogen target remains retrospective; frequency and reciprocal period are two reports of the same comparison.

Provenance. `manual2/data/m1/atomic/` stores separate calculation and comparison records and the construction obligations. The frozen inventory remains under `manual2/data/m1/hyperfine/`.

Literature note. Defining scale: BIPM [1]. Historical target: Hellwig et al. [3]. Hyperfine Hamiltonian: Steck [17]. Magnetic conversion: NIST [9].

8.3 Elapsed seconds and a joint uncertainty budget

Setup. For a committed reference window ω, let C_r be the physical cycle accumulation established by an observation map, f_0 the fixed nominal reference frequency, $\bar{y}$ its effective fractional offset over that window, and d the signed correction from observed to intended endpoints. Native reclosure count and physical oscillator count are distinct inputs. All shared calibration parameters retain their covariance.

Derivation. The declared measurement equation and its first-order sensitivities are

$$\widehat{T}(\omega) = \frac{C_r}{f_0(1+\bar{y})} + d, \tag{22}$$

$$J = \left(\frac{1}{f_0(1+\bar{y})}, -\frac{C_r}{f_0(1+\bar{y})^2}, 1\right), \qquad u_c^2(T) = J\Sigma J^{\mathsf{T}}. \tag{23}$$

The elapsed interval $\widehat{T}(\omega)$ is distinct from the transition period $\widehat{T}_{\mathrm{H}}$ in (19). A faster reference gives a shorter inferred interval at fixed accumulated cycles. An ideal frequency divider transforms counts, nominal frequency and covariance together and preserves the reported interval and its uncertainty.

Specialization. A shared endpoint offset cancels only when the measurement equation takes its difference. A common frequency bias need not average down. A quadratic field shift at zero mean field can have nonzero expectation and uncertainty even where its first derivative vanishes. Unresolved discrete cycle slips require an ambiguity model. Unknown covariance, drift or transfer effects are not numerical zeros.

Calibration. Each budget component records its unit, source, validity window, correction sign, covariance group and inclusion in any aggregate. An already included realization contribution is not added again as an independent term. The reported measurand distinguishes local proper time from a coordinated time scale; only applicable external motion, potential and transfer corrections enter. A later calibration may revise a retrospective report, but it cannot be represented as available to an earlier online computation.

The frozen v2 records retain first-order diagnostic budgets and their original one-percent marginal guards. Those guards do not prove nonlinear adequacy: the M3 successor checks exhibit a near-cancelled product whose exact variance is 50.90005 times its first-order value, and a zero-derivative interval with nonzero exact variance. Main (168) supplies full finite-law propagation when that joint law is supplied; covariance alone does not complete a nonlinear model.

The persisted positive examples exercise declared synthetic maps and uncertainty budgets. They do not invent a clock calibration certificate. A materially incomplete physical budget produces a scoped partial result; independent finite-model execution continues.

One exact synthetic budget yields $\widehat{T} = 1\,\mathrm{s}$ and $u_c^2 = 4904\,\mathrm{ps}^2$, hence $u_c \simeq 70.0286\,\mathrm{ps}$. A separate integration example maps five reference reclosures to one second by stipulation; the controlled intervention changes the compared count from six to two while preserving those five reference reclosures. Neither stipulation supplies a measured atomic observation gain.

Falsification. The integration checks reject covariance with incorrect units, ordering, asymmetry or a negative quadratic direction; unknown material terms, duplicate aggregate inclusion, stale scope and future calibration are tested separately. Nonlinear and discrete cases cannot acquire a complete precision claim by using only a linear derivative.

Provenance. `manual2/data/m1/metrology/`; the application contract is preserved in `manual2/evidence/v2-request/SI_ANCHOR_CONTRACT.md`.

Literature note. Realization and proper time: CCTF [2]. Measurement models and correlated uncertainty: JCGM [4, 5].

8.4 Published frequency covariance

Setup. The BIPM adjustment input retains 153 rows, including seven effectively excluded by enlarged uncertainties, and 1,168 correlation coefficients. The output contains 14 frequency estimates and their covariance. Raw bytes, source metadata and prior target exposure are preserved.

Derivation. The source README describes a ratio basis, while the numerical matrix is consistent with squared hertz. Under this documented interpretation, ratios to the exact Cs definition use $\Sigma_{\nu/\nu_{\mathrm{Cs}}} = \Sigma_\nu/\nu_{\mathrm{Cs}}^2$; general ratios use their full Jacobian. Exact rational intervals propagate the nearest-rounding cells of covariance inputs and printed frequencies. All 105 calculated uncertainty intervals overlap the

corresponding squared output-uncertainty rounding cells. This includes cases where a point estimate alone lies outside the last-digit interval of the output. The preserved check used a nearest-rounding cell for every token. The separate M3 precision successor identifies 26 printed-zero cells affecting 13 ratios: their formatting or underflow convention is not supplied by the source. Treating those zeros as point transcriptions is a conditional numerical check, not a source-exact uncertainty claim. All 105 point results are preserved; a one-percent sigma mutation in the selected affected ratio is rejected under that conditional policy.

Specialization. The recommendation multiplies standard uncertainty by 2.35, hence covariance by 2.35^2. This is distinct from a coverage factor. The adjustment's added fractional term $5.324065020 \times 10^{-17}$ is already represented in its outputs and is not appended again. Adjusted estimates and their contributing measurements are not independent observations of one another.

Falsification. Hash checks preserve the input; independent parser and numerical tests reject altered identities, correlation domains, covariance, units and output uncertainties. The numerical unit interpretation is not an official corrigendum, a reconstruction of the least-squares adjustment or a physical clock realization.

Provenance. `manual2/data/m1/bipm/source_manifest.json` , `bipm_report.json` and the separate `bipm_precision_report.json`; 153 source records and all 105 uncertainty comparisons remain inspectable.

Literature note. Published input, output, covariance and analysis notes: BIPM [7].

9 Isotope Half-Lives and a Common Survival Experiment

9.1 Source records and nucleon bookkeeping

Setup. The element reference remains the 118 placements and 126 slots of §3. A separate isotope layer joins by atomic number and symbol. It retains the nuclear-state assignment, selected evaluation, original notation, asymmetric uncertainty, decay channels and experimental overlap. Total half-life, channel-partial half-life and an individual waiting time are distinct quantities. Observational stability supplies no measured infinity.

Derivation. For the formal grouping of a record with proton count Z, neutron count $N = A - Z$, define

$$q = \min(\lfloor Z/2 \rfloor, \lfloor N/2 \rfloor), \quad r_p = Z - 2q, \quad r_n = N - 2q, \qquad A = 4q + r_p + r_n. \tag{24}$$

These are exact integer coordinates. Identifying a group with a physical alpha cluster requires an additional construction. The bookkeeping alone supplies neither a loss probability nor a half-life.

Specialization. Tables 6–8 keep the printed reference compact. The companion retains the broader family records, source-specific updates and uncertainty qualifications. H-1 and Cs-133 retain their recurrence roles. Fe-56 is the selected binding benchmark; the Ni-62 digital control prevents calling Fe-56 the maximum binding energy per nucleon. Pb-208 remains observationally stable, with its experimental alpha-channel limit stored separately from total lifetime.

Isotope	Z, N	q	r_p, r_n	Adopted $T_{1/2}$
^{1}H	$1, 0$	0	$1, 0$	observationally stable
^{133}Cs	$55, 78$	27	$1, 24$	observationally stable

Table 6: Recurrence anchors. Stability is an observational classification; no infinite half-life is assigned.

Isotope	Z, N	q	r_p, r_n	Adopted $T_{1/2}$
^{56}Fe	$26, 30$	13	$0, 4$	observationally stable
^{208}Pb	$82, 126$	41	$0, 44$	observationally stable

Table 7: Binding and observational-stability controls. The digital reference retains Ni-62 and the channel-specific Pb-208 bound.

Isotope	Z, N	q	r_p, r_n	Adopted $T_{1/2}$
^{294}Og	$118, 176$	59	$0, 58$	$0.58^{+0.44}_{-0.18}$ ms
^{290}Lv	$116, 174$	58	$0, 58$	$8.3^{+3.5}_{-1.9}$ ms
^{286}Fl	$114, 172$	57	$0, 58$	166^{+40}_{-27} ms
^{293}Lv	$116, 177$	58	$0, 61$	95^{+63}_{-27} ms
^{289}Fl	$114, 175$	57	$0, 61$	$2.4^{+0.8}_{-0.5}$ s
^{298}Fl	$114, 184$	57	$0, 70$	unobserved target

Table 8: Selected superheavy records. Asymmetric uncertainties are expanded from the adopted last-digit notation. Fl-298 is an unobserved target. Newer experimental analyses remain separate in the companion.

$$^{294}\mathrm{Og} \xrightarrow{\alpha} {}^{290}\mathrm{Lv} \xrightarrow{\alpha} {}^{286}\mathrm{Fl} \xrightarrow{\alpha} {}^{282}\mathrm{Cn} \xrightarrow{\mathrm{SF}} \text{fragments}$$
$$^{293}\mathrm{Lv} \xrightarrow{\alpha} {}^{289}\mathrm{Fl} \xrightarrow{\alpha} {}^{285}\mathrm{Cn}$$

Figure 2: Observed alpha relationships in the selected records. The first route is one branch: Fl-286 also fissions. Cn-285 has further alpha decay; it is a scope endpoint here, not an asserted terminal stable state.

Calibration. Evaluated half-lives retain their source units and explicit conversion to seconds. Newer experimental analyses remain distinct from the adopted evaluation; overlapping events are not averaged as independent data. Fl-298 has no measured half-life and receives no empirical score. The progression Fl-289, Og-294, Fl-298 is a comparison set, not a decay chain. The sourced alpha edges in Fig. 2 obey $\Delta Z = -2$, $\Delta A = -4$; spontaneous fission does not acquire an invented unique daughter.

Falsification. Source, state, quantity and status mutations must fail without rewriting inputs. The logarithmic half-life display names the selected isotope and gives limits, stability, unknown records and uncovered elements nonnumeric treatments. It does not reinterpret the structural-role colors or an arbitrary A arithmetic input as measured stability.

Provenance. `manual2/data/m1/isotopes/` binds the source bytes, selected records, updates and generated tables. The original element ledger and historical numeric records remain unchanged.

Literature note. Evaluated isotope data: ENSDF [12]; evaluation cross-check: NUBASE2020 [6]; binding control: AME2020 [19].

9.2 A state-dependent finite law

Setup. The third application model instantiates the conditional H1 kernel of Main (66), paired orientation response of Main (96), and excess construction of Main (109). Main does not select a nuclear state, its kernel weights or a physical loss observation for this example. Within the declared finite scope, the application fixes two retained square paths

$$\gamma_+ = (0000, 1000, 1001, 0001, 0000), \tag{25}$$

$$\gamma_- = (0000, 0100, 0101, 0001, 0000). \tag{26}$$

Their vertices have one injective Q4 encoding. At 0000, the incoming ancestor slot is 0001; the three continuations are $1000, 0100, 0010$. The third exits through 0010, then takes 1010 or 0110. These last routes carry hypothetical alpha and fission labels, each with declared conditional weight one. Their nuclear identification is not established.

A separate reference lineage repeats γ_+. Both lineages begin with two distinct retained-start witnesses. One step of the survival experiment completes a new reference return, binds the already committed candidate prefix, normalizes the kernel, executes the selected path and commits the response for the next contact. Four H1 traversals and the retained start witness the finite square return; passing this predicate does not establish native motif admission. The structural hinge is retained. No universal seconds per traversal or scheduler cut are assigned.

Derivation. Let h be a particular retained choice history, with signed committed memory R_h, common capacity $C \geq 0$, and common positive loss gain λ. The declared specialization is

$$X_h = \max(0, |R_h| - C), \qquad (w_+, w_-, w_{\text{exit}}) = (1, 1, \lambda X_h), \tag{27}$$

$$P_+(h) = P_-(h) = \frac{1}{2 + \lambda X_h}, \qquad P_{\text{exit}}(h) = \frac{\lambda X_h}{2 + \lambda X_h}, \tag{28}$$

$$R_{h\sigma} = R_h + \sigma, \qquad \sigma \in \{-1, +1\}. \tag{29}$$

The last update retains the oriented response pair $(\sigma, -\sigma)$. Its unit magnitude specializes declared pressure 2 and signed lag magnitude $1/2$; neither is an isotope-derived support accessor or a lag inferred

from measured decay. This is the explicit application premise between the source response form and the weight law. Zero exit weight leaves the exit structurally admitted and unrealized.

For exact history mass μ_h, multiplication by the probabilities in (28) supplies the child masses. Absorbed histories retain their completed return IDs and inner scope map. They have no later candidate return. Distinct prefixes remain distinct even if their numerical memories agree. Summing only retained histories after reference return c gives

$$S(c) = \Pr(C_{\text{loss}} > c) = \sum_{h \text{ retained after } c} \mu_h, \qquad c_{1/2} = \min\{c : S(c) \le \tfrac{1}{2}\}. \tag{30}$$

Thus $c_{1/2}$ is the first discrete crossing. The recorded crossing interval $(c_{1/2} - 1, c_{1/2}]$ makes no within-cycle interpolation. If no crossing occurs by the horizon, the result is a censored lower bound. A single loss event is not this ensemble median.

Specialization. With $R_\varnothing = 0, C = 1, \lambda = 1$, exact propagation gives

$$\big(S(0), \ldots, S(8)\big) = \left(1, 1, 1, \frac{5}{6}, \frac{3}{4}, \frac{3}{5}, \frac{19}{36}, \frac{703}{1680}, \frac{211}{576}\right). \tag{31}$$

The median crosses in $(6, 7]$ reference returns. Each run expands 255 retained prefixes without merging. Doubling the common loss gain gives $(4, 5]$; setting the common capacity to zero gives $(2, 3]$. Disabling new response commits at return 2 leaves $S(c) = 1$ through return 8. Its survival agrees with the baseline through return 2 and first differs at return 3: 1 versus 5/6. This controlled change therefore affects subsequent loss rather than merely annotating a precomputed trace.

The finite law is evaluated with the exact M3 mapping implementation and Main (168). For the retained indicator at return 8, its expectation is 211/576 and its variance is (211/576)(365/576). This ensemble spread is distinct from uncertainty about the nuclear model or the experimental calibration.

Calibration. No isotope-specific hazard or capacity is fitted. Every named isotope currently labels the same surrogate configuration; (24) does not change its law. A physical elapsed-time report would require an observation map taking reference returns to observed oscillator cycles. For example, the hypothetical constant gain $g > 0$ would give $T = gc/\Delta\nu_{\text{Cs}}$, with $\Delta\nu_{\text{Cs}} = 9\,192\,631\,770\,\text{Hz}$. The unselected gains $g = 1, 2$ give different times while preserving all native histories and the same SI definition. The defining frequency alone therefore cannot select a decay-time prediction. Realization, model discrepancy and covariance remain unresolved rather than numerical zeros.

Falsification. The exact tests reject probability leakage, coordinate aliases, lost lineage, a forged return, nonantisymmetric response, later survival disconnected from feedback, and status promotion. The target-free prediction runs without the isotope catalog. The comparison replays that complete prediction before consuming validated source records. A changed target or a self-consistent replacement hash cannot rewrite the frozen prediction.

Provenance. `manual2/data/m1/survival/` records the common model contract, four exact runs, prediction, comparison and unresolved construction obligations. The earlier finite folding and stateful fixtures remain separate. The observation-gain dependency instantiates Main §7.5.

Literature note. SI reference definition: BIPM [1].

9.3 A measured test of the restricted specialization

Setup. Equal surrogate states with a common observation and clock map imply identical survival distributions and a half-life ratio of one. This restricted consequence is testable even though the isotope-to-native-state construction is incomplete. The selected adopted records were exposed before this calculation; this is a retrospective comparison.

Derivation. Using numerator and denominator source uncertainty envelopes without treating them as independent Gaussian inputs yields

Half-life ratio	Adopted central ratio	Source-envelope ratio range
Og-294 / Lv-290	29/415	$[2/59, 51/320]$
Lv-290 / Fl-286	1/20	$[16/515, 59/695]$
Lv-293 / Fl-289	19/480	$[17/800, 79/950]$

All three ranges exclude one. These are descriptive enclosures constructed from the reported uncertainties, not a joint confidence statement or significance score.

Specialization. The identity-insensitive, shared-clock specialization disagrees with these reported envelopes. A successful common law must therefore distinguish the isotope configurations or their justified observation relations. The measurements do not select a replacement law. This result does not falsify the general AOD construction or establish a native nuclear prediction. The next obligations are an admitted isotope/state configuration, an independently justified state-to-loss law, the reference observation map and a complete physical uncertainty model. Fl-298 remains an unscored target.

Calibration. A separate conventional benchmark imports each adopted $T_{1/2}$ and defines $S(t) = \exp(-\lambda_{\mathrm{exp}} t)$, $\lambda_{\mathrm{exp}} = \ln 2 / T_{1/2}$. Its mean lifetime is $T_{1/2}/\ln 2$, not $T_{1/2}$. Reproducing survival 1/2 at an imported half-life checks the conventional calculation; it does not predict that half-life. The native finite law in (28) is not assumed exponential.

Falsification. A physical isotope prediction must fix its state construction and observation law before joining the evaluation target. Independent prediction additionally requires an exposure design beyond these already known records. Source-specific revisions remain separately bound comparisons, and no unobserved lifetime can be counted as agreement.

Provenance. The separate `prediction.json` and `comparison.json` bind their complete inputs and exact outputs. Tests SURV-01–19 state their finite assertion scope; passing them does not certify nuclear admission.

10 Reference, History and Pending-Comparison Registry

Setup. This registry distinguishes active static calculations, a declared finite model, external source records, preserved historical declarations and pending native or measured comparisons. It does not classify every integer table as a completed native simulation.

Table 9: **Matter I evidence inventory.**

Record	Present scope	Completion boundary
Support polynomial	Exact integer expression	Does not admit a motif
Enclosure groups and local tuples	Exact grouping and role projection	Does not execute a path
Reference-A and assigned classes	Pinned candidate fixture inputs	Original selection provenance unresolved
Conditional support and residuals	Exact consequences of those inputs	Native accessor/retention claims pending
v94 placement crosswalk	118 rows, six compared fields	708 comparisons; separate from lane inventory
v94 observable inventory	118 elements across E0–E5	Preserve original per-lane statuses
r24 occurrence/packet crosswalk	Historical declaration identities	Native transition and SADAR execution deferred
Element-wide D.E.C. and certificates	Required schema and obligations	Not executed; admission pending
Two-face construction	Injective Q4 incidence; external event ledger	Illustrative traversal count only
Finite fold and recurrence v1	Mapped histories and calculated responses	Frozen declared fixture; native admission pending
Sequential recurrence v2	Response changes later retention, returns and contacts	Declared state dependence executed
SI observation layer	Synthetic maps, covariance and uncertainty cases	No device calibration supplied
BIPM adjustment data	Raw records and 105 covariance checks	Units inference explicit; adjustment not reproduced
Cs/H experiment	Conventional benchmark and observation-map countermodels	AOD atomic prediction underdetermined
Hydrogen/caesium inventory	Two isotopes, four states, two transitions; historical ratio	Source inventory verified; physical map and scores pending
Isotope catalog	Source-qualified Og/Lv/Fl families, anchors and controls	External observations; tentative and unobserved states explicit
Survival ensemble	Exact history-dependent retained/loss law	Median in reference returns; isotope map and SI gain unresolved
M3 mapping and BIPM precision	Typed finite-law propagation and separate zero-policy check	Nonlinear adequacy and zero precision remain scoped
Other mass and spectral lanes	Future sourced inputs and comparisons	No element-wide scores supplied

Specialization. The original v94 observable matrix has 708 rows: 118 E0 identity/registry rows marked `active_registry`, and 118 rows in each of E1 placement/classification, E2 atomic mass/standard atomic weight, E3 isotope support, E4 spectra and E5 reaction/routes marked `declared_lane_not_scored_in_v94`. Its ten value maps remain `deferred_not_attached`. These statuses are separate from the successful six-field placement comparison.

The r24 crosswalk retains occurrence, transition-packet and SADAR declaration IDs as historical IDs.

Its execution remains declared but unexecuted; target values are not read, measured reports are closed and scores absent. It does not contain source-to-destination event witnesses. Correspondence of the original and public atlas is bound by parsed manifest content and five matching file hashes; it is not an identity claim about an entire public repository.

Falsification. A historical pending lane promoted to a current execution result without new evidence, or a hash comparison described as scientific admission, violates the inventory. Preserve predecessor status and current status in distinct fields. Prerecorded counterfactual outcomes do not count as actual hostile tests.

Provenance. The release provenance records the reviewed PR, original v94/r24 archive hashes, extracted member hashes, the six-field mapping, historical identifier joins and the current input bindings. Small verified source extracts and the generated `declaration_crosswalk.csv`, `observable_coverage.csv` and `value_map_status.csv` are stored under `manual2/data/m1/history/`. Their bindings are in `manual2/evidence/historical_sources.json`; current input hashes are in `manual2/evidence/input_manifest.json`. Exact static checks, source identity checks and any future scientific claims retain separate result scopes.

A Exact Arithmetic Fixtures

A.1 Scope

These are arithmetic fixtures. They do not claim simulated native trajectories or certified closure. The inherited four-step convention is in Main App. I; polynomial support is in Main App. H.

A.2 Four-step decompositions

For $L = 4c_L + r_L$, $r_L \in \{0, 1, 2, 3\}$, the outer even indices give

$$\begin{aligned} 6 &= 4(1) + 2, & 8 &= 4(2) + 0, & 10 &= 4(2) + 2, \\ 12 &= 4(3) + 0, & 14 &= 4(3) + 2, & 16 &= 4(4) + 0, \\ 18 &= 4(4) + 2, & 20 &= 4(5) + 0, & 22 &= 4(5) + 2. \end{aligned}$$

Even L has remainder 0 or 2; an odd index has remainder 1 or 3. This division identity is not a theorem that every closure has four steps. The inherited history-weighted load requires its declared histories and weights (Main (181)).

A.3 Labelled-reference occupancy

The 126 declared slots and 118 element labels give

$$\frac{118}{126} = \frac{59}{63}, \qquad \frac{N_{\mathrm{labelled}}(\mathsf{C}_{22})}{22} = \frac{14}{22} = \frac{7}{11}, \qquad \frac{N_{\mathrm{unfilled}}(\mathsf{C}_{22})}{22} = \frac{8}{22} = \frac{4}{11}.$$

These are exact occupancy fractions, independent of assigned TTL values and native retention.

A.4 Falsification

Non-integer quotient/remainder, an invalid remainder, or disagreement with the labelled-slot counts fails the corresponding arithmetic fixture. It does not by itself falsify a native execution that has not been performed.

Provenance. Main App. I.2, Main App. I.4, (10).

B Pending External Comparison

B.1 Scope and numeric policy

No native-versus-measured atomic score is executed in this release. The external inventory in §7 records a historical hydrogen measurement and the caesium defining reference; reference arithmetic is separate from a model prediction. The diagnostic residuals in §8 compare unselected observation-map countermodels with the exposed target; they do not establish an atomic model or agreement. An atomic comparison must bind an established candidate, its declared value map and a sourced target record before evaluating

$$\mathsf{Q}_{\mathrm{ext}}(\nu_{Z,A\leftrightarrow\Omega};\mathcal{T}_Z). \tag{32}$$

A conditional support sum does not become a measured prediction merely by being frozen. Candidate selection and the mathematical/physical meaning of the value map must be explicit before joining targets.

External mass, spectral or isotope data retain their sources and units. A displayed measured decimal requires

$$(x_{\mathrm{disp}}, u_x, \mathrm{unit}(x), \mathrm{src}(x), \mathrm{round}(x), \mathrm{scope}(x)). \tag{33}$$

These records are distinct from the exact integer reference A and from $(Z, A-Z)$. AME and NIST references below identify possible comparison sources; no per-element selection of the present A inputs is attributed to them.

B.2 Pending comparison gates

A future measured lane must identify the candidate and frozen inputs, applicable admission/execution evidence, value map, target version and row identity, units and precision, comparison formula, and resulting status. Missing inputs remain explicitly pending. No spectral, isotope-stability, charge, decay or reaction agreement is inferred from the reference figure.

Provenance. Main §5.16.12; released AOD Manual v41.02, §1.7–§1.8 and App. B.

Literature note. Potential external comparison sources: Wang et al. [18]; NIST ASD [11].

C Exact Parity, Role and Occupancy Comparators

C.1 Integer comparators

On each complete reference group C_{2m}, $m = 3, \ldots, 11$:

1. Exactly m seats have odd local parity and m even local parity. Against the comparator $\widehat{O}_m = m$,

$$\Delta^{\mathbb{Z}}_{\text{parity},m} = \widehat{O}_m - O_{\text{even},m} = 0.$$

   Parity does not determine the assigned operational class.

2. The declared role projection has $m-1$ first-branch seats, one hinge seat, $m-1$ counterbranch seats and one boundary seat. Thus

$$\Delta^{\mathbb{Z}}_{\text{branch},m} = (m-1) - (m-1) = 0.$$

   These are counts of visual roles, not traversed edge events.

3. The terminal group has 14 labelled slots and capacity 22:

$$\Delta^{\mathbb{Z}}_{\mathsf{C}_{22}} = 14 - 22 = -8.$$

   The difference counts eight unfilled reference slots and is not a retention or physical-cutoff residual.

C.2 Signed-magnitude record

The comparator representation is

$$\Delta^{\mathbb{Z}}_k = s^{(3)}_k m_k, \qquad s^{(3)}_k \in \{-1, 0, +1\}, \quad m_k = |\Delta^{\mathbb{Z}}_k|.$$

The two balanced comparisons give $\delta_3 = (0, 0)$; the terminal occupancy comparison gives $\delta_3 = (-1, 8)$. These values express the declared integer counts, not a native phase-cycle measurement.

Provenance. (11), (12), Main §5.7.13; released AOD Manual v41.02, App. C, for the signed comparator convention.

D Complete Element Reference Ledger

The following table preserves the full declared reference. Its numerical and historical fields are unchanged by the isotope and survival extension.

Table 10: **Complete declared reference and conditional support ledger.** $Z = 1 \ldots 118$ names supplied elements; slots $119 \ldots 126$ are unfilled reference positions. Pressure and route classes are assigned inputs; the support sum is a conditional expression, not an admitted native duration. Grouping follows Main §5.7.13; the expression specializes Main (35).

Z	Sym	Ref.	A	C_{2m}	Local $a{:}b{:}c$	u_-	h	u_+	U	P^{fix}	C^{fix}	$S^{\text{fix}}, X^{\text{fix}}$	Assigned class
1	H	1		C_6	1 : 0 : 1	30	1	4	35	15	18	$-3, 0$	Deficit
2	He	4		C_6	1 : 1 : 0	196	1	8	205	18	18	$0, 0$	Neutral
3	Li	7		C_6	2 : 1 : 1	362	1	12	375	18	18	$0, 0$	Neutral
4	Be	9		C_6	2 : 2 : 0	460	1	16	477	18	18	$0, 0$	Neutral
5	B	11		C_6	3 : 2 : 1	558	1	20	579	18	18	$0, 0$	Neutral
6	C	12		C_6	3 : 3 : 0	588	1	24	613	18	18	$0, 0$	Neutral + anchor
7	N	14		C_8	1 : 0 : 1	686	1	4	691	24	24	$0, 0$	Neutral
8	O	16		C_8	1 : 1 : 0	784	1	8	793	21	24	$-3, 0$	Deficit
9	F	19		C_8	2 : 1 : 1	950	1	12	963	21	24	$-3, 0$	Deficit
10	Ne	20		C_8	2 : 2 : 0	980	1	16	997	24	24	$0, 0$	Neutral
11	Na	23		C_8	3 : 2 : 1	1146	1	20	1167	24	24	$0, 0$	Neutral
12	Mg	24		C_8	3 : 3 : 0	1176	1	24	1201	24	24	$0, 0$	Neutral

Z	Sym	Ref.	A	C_{2m}	Local $a{:}b{:}c$	u_-	h	u_+	U	P^{fix}	C^{fix}	$S^{\mathrm{fix}}, X^{\mathrm{fix}}$	Assigned class
13	Al	27		C_8	4 : 3 : 1	1342	1	28	1371	24	24	0, 0	Neutral
14	Si	28		C_8	4 : 4 : 0	1372	1	32	1405	24	24	0, 0	Neutral + anchor
15	P	31		C_{10}	1 : 0 : 1	1538	1	4	1543	30	30	0, 0	Neutral
16	S	32		C_{10}	1 : 1 : 0	1568	1	8	1577	27	30	−3, 0	Deficit
17	Cl	35		C_{10}	2 : 1 : 1	1734	1	12	1747	27	30	−3, 0	Deficit
18	Ar	40		C_{10}	2 : 2 : 0	2036	1	16	2053	30	30	0, 0	Neutral
19	K	39		C_{10}	3 : 2 : 1	1930	1	20	1951	30	30	0, 0	Neutral
20	Ca	40		C_{10}	3 : 3 : 0	1960	1	24	1985	30	30	0, 0	Neutral
21	Sc	45		C_{10}	4 : 3 : 1	2262	1	28	2291	30	30	0, 0	Neutral
22	Ti	48		C_{10}	4 : 4 : 0	2428	1	32	2461	30	30	0, 0	Neutral
23	V	51		C_{10}	5 : 4 : 1	2594	1	36	2631	30	30	0, 0	Neutral
24	Cr	52		C_{10}	5 : 5 : 0	2624	1	40	2665	30	30	0, 0	Neutral + anchor
25	Mn	55		C_{12}	1 : 0 : 1	2790	1	4	2795	36	36	0, 0	Neutral
26	Fe	56		C_{12}	1 : 1 : 0	2820	1	8	2829	36	36	0, 0	Neutral
27	Co	59		C_{12}	2 : 1 : 1	2986	1	12	2999	36	36	0, 0	Neutral
28	Ni	58		C_{12}	2 : 2 : 0	2880	1	16	2897	36	36	0, 0	Neutral
29	Cu	63		C_{12}	3 : 2 : 1	3182	1	20	3203	36	36	0, 0	Neutral
30	Zn	64		C_{12}	3 : 3 : 0	3212	1	24	3237	36	36	0, 0	Neutral
31	Ga	69		C_{12}	4 : 3 : 1	3514	1	28	3543	36	36	0, 0	Neutral
32	Ge	74		C_{12}	4 : 4 : 0	3816	1	32	3849	36	36	0, 0	Neutral
33	As	75		C_{12}	5 : 4 : 1	3846	1	36	3883	36	36	0, 0	Neutral
34	Se	80		C_{12}	5 : 5 : 0	4148	1	40	4189	33	36	−3, 0	Deficit
35	Br	79		C_{12}	6 : 5 : 1	4042	1	44	4087	33	36	−3, 0	Deficit
36	Kr	84		C_{12}	6 : 6 : 0	4344	1	48	4393	36	36	0, 0	Neutral + anchor
37	Rb	85		C_{14}	1 : 0 : 1	4374	1	4	4379	42	42	0, 0	Neutral
38	Sr	88		C_{14}	1 : 1 : 0	4540	1	8	4549	42	42	0, 0	Neutral
39	Y	89		C_{14}	2 : 1 : 1	4570	1	12	4583	42	42	0, 0	Neutral
40	Zr	90		C_{14}	2 : 2 : 0	4600	1	16	4617	42	42	0, 0	Neutral
41	Nb	93		C_{14}	3 : 2 : 1	4766	1	20	4787	42	42	0, 0	Neutral
42	Mo	98		C_{14}	3 : 3 : 0	5068	1	24	5093	42	42	0, 0	Neutral
43	Tc	98		C_{14}	4 : 3 : 1	5030	1	28	5059	45	42	3, 3	Surplus
44	Ru	102		C_{14}	4 : 4 : 0	5264	1	32	5297	42	42	0, 0	Neutral
45	Rh	103		C_{14}	5 : 4 : 1	5294	1	36	5331	42	42	0, 0	Neutral
46	Pd	106		C_{14}	5 : 5 : 0	5460	1	40	5501	42	42	0, 0	Neutral
47	Ag	107		C_{14}	6 : 5 : 1	5490	1	44	5535	42	42	0, 0	Neutral
48	Cd	114		C_{14}	6 : 6 : 0	5928	1	48	5977	42	42	0, 0	Neutral
49	In	115		C_{14}	7 : 6 : 1	5958	1	52	6011	42	42	0, 0	Neutral
50	Sn	120		C_{14}	7 : 7 : 0	6260	1	56	6317	42	42	0, 0	Neutral + anchor
51	Sb	121		C_{16}	1 : 0 : 1	6290	1	4	6295	48	48	0, 0	Neutral
52	Te	130		C_{16}	1 : 1 : 0	6864	1	8	6873	45	48	−3, 0	Deficit
53	I	127		C_{16}	2 : 1 : 1	6622	1	12	6635	45	48	−3, 0	Deficit
54	Xe	132		C_{16}	2 : 2 : 0	6924	1	16	6941	48	48	0, 0	Neutral
55	Cs	133		C_{16}	3 : 2 : 1	6954	1	20	6975	48	48	0, 0	Neutral
56	Ba	138		C_{16}	3 : 3 : 0	7256	1	24	7281	48	48	0, 0	Neutral
57	La	139		C_{16}	4 : 3 : 1	7286	1	28	7315	48	48	0, 0	Neutral
58	Ce	140		C_{16}	4 : 4 : 0	7316	1	32	7349	48	48	0, 0	Neutral
59	Pr	141		C_{16}	5 : 4 : 1	7346	1	36	7383	48	48	0, 0	Neutral
60	Nd	142		C_{16}	5 : 5 : 0	7376	1	40	7417	48	48	0, 0	Neutral
61	Pm	145		C_{16}	6 : 5 : 1	7542	1	44	7587	51	48	3, 3	Surplus
62	Sm	152		C_{16}	6 : 6 : 0	7980	1	48	8029	48	48	0, 0	Neutral
63	Eu	153		C_{16}	7 : 6 : 1	8010	1	52	8063	48	48	0, 0	Neutral
64	Gd	158		C_{16}	7 : 7 : 0	8312	1	56	8369	48	48	0, 0	Neutral
65	Tb	159		C_{16}	8 : 7 : 1	8342	1	60	8403	48	48	0, 0	Neutral
66	Dy	164		C_{16}	8 : 8 : 0	8644	1	64	8709	48	48	0, 0	Neutral + anchor
67	Ho	165		C_{18}	1 : 0 : 1	8674	1	4	8679	54	54	0, 0	Neutral
68	Er	166		C_{18}	1 : 1 : 0	8704	1	8	8713	54	54	0, 0	Neutral
69	Tm	169		C_{18}	2 : 1 : 1	8870	1	12	8883	54	54	0, 0	Neutral
70	Yb	174		C_{18}	2 : 2 : 0	9172	1	16	9189	54	54	0, 0	Neutral
71	Lu	175		C_{18}	3 : 2 : 1	9202	1	20	9223	54	54	0, 0	Neutral
72	Hf	180		C_{18}	3 : 3 : 0	9504	1	24	9529	54	54	0, 0	Neutral
73	Ta	181		C_{18}	4 : 3 : 1	9534	1	28	9563	54	54	0, 0	Neutral
74	W	184		C_{18}	4 : 4 : 0	9700	1	32	9733	54	54	0, 0	Neutral
75	Re	187		C_{18}	5 : 4 : 1	9866	1	36	9903	54	54	0, 0	Neutral
76	Os	192		C_{18}	5 : 5 : 0	10168	1	40	10209	54	54	0, 0	Neutral
77	Ir	193		C_{18}	6 : 5 : 1	10198	1	44	10243	54	54	0, 0	Neutral
78	Pt	195		C_{18}	6 : 6 : 0	10296	1	48	10345	54	54	0, 0	Neutral
79	Au	197		C_{18}	7 : 6 : 1	10394	1	52	10447	54	54	0, 0	Neutral
80	Hg	202		C_{18}	7 : 7 : 0	10696	1	56	10753	54	54	0, 0	Neutral

Z	Sym	Ref.	A	C_{2m}	Local $a{:}b{:}c$	u_-	h	u_+	U	P^{fix}	C^{fix}	$S^{\mathrm{fix}}, X^{\mathrm{fix}}$	Assigned class
81	Tl	205		C_{18}	8 : 7 : 1	10862	1	60	10923	54	54	0, 0	Neutral
82	Pb	208		C_{18}	8 : 8 : 0	11028	1	64	11093	54	54	0, 0	Neutral
83	Bi	209		C_{18}	9 : 8 : 1	11058	1	68	11127	54	54	0, 0	Neutral
84	Po	209		C_{18}	9 : 9 : 0	11020	1	72	11093	57	54	3, 3	Surplus + anchor
85	At	210		C_{20}	1 : 0 : 1	11050	1	4	11055	63	60	3, 3	Surplus
86	Rn	222		C_{20}	1 : 1 : 0	11828	1	8	11837	63	60	3, 3	Surplus
87	Fr	223		C_{20}	2 : 1 : 1	11858	1	12	11871	63	60	3, 3	Surplus
88	Ra	226		C_{20}	2 : 2 : 0	12024	1	16	12041	63	60	3, 3	Surplus
89	Ac	227		C_{20}	3 : 2 : 1	12054	1	20	12075	63	60	3, 3	Surplus
90	Th	232		C_{20}	3 : 3 : 0	12356	1	24	12381	66	60	6, 6	Surplus
91	Pa	231		C_{20}	4 : 3 : 1	12250	1	28	12279	66	60	6, 6	Surplus
92	U	238		C_{20}	4 : 4 : 0	12688	1	32	12721	66	60	6, 6	Surplus
93	Np	237		C_{20}	5 : 4 : 1	12582	1	36	12619	66	60	6, 6	Surplus
94	Pu	244		C_{20}	5 : 5 : 0	13020	1	40	13061	66	60	6, 6	Surplus
95	Am	243		C_{20}	6 : 5 : 1	12914	1	44	12959	66	60	6, 6	Surplus
96	Cm	247		C_{20}	6 : 6 : 0	13148	1	48	13197	69	60	9, 9	Surplus
97	Bk	247		C_{20}	7 : 6 : 1	13110	1	52	13163	69	60	9, 9	Surplus
98	Cf	251		C_{20}	7 : 7 : 0	13344	1	56	13401	69	60	9, 9	Surplus
99	Es	252		C_{20}	8 : 7 : 1	13374	1	60	13435	69	60	9, 9	Surplus
100	Fm	257		C_{20}	8 : 8 : 0	13676	1	64	13741	69	60	9, 9	Surplus
101	Md	258		C_{20}	9 : 8 : 1	13706	1	68	13775	69	60	9, 9	Surplus
102	No	259		C_{20}	9 : 9 : 0	13736	1	72	13809	72	60	12, 12	Surplus
103	Lr	266		C_{20}	10 : 9 : 1	14174	1	76	14251	72	60	12, 12	Surplus
104	Rf	267		C_{20}	10 : 10 : 0	14204	1	80	14285	72	60	12, 12	Surplus + anchor
105	Db	268		C_{22}	1 : 0 : 1	14234	1	4	14239	78	66	12, 12	Surplus
106	Sg	269		C_{22}	1 : 1 : 0	14264	1	8	14273	78	66	12, 12	Surplus
107	Bh	270		C_{22}	2 : 1 : 1	14294	1	12	14307	78	66	12, 12	Surplus
108	Hs	277		C_{22}	2 : 2 : 0	14732	1	16	14749	81	66	15, 15	Surplus
109	Mt	278		C_{22}	3 : 2 : 1	14762	1	20	14783	81	66	15, 15	Surplus
110	Ds	281		C_{22}	3 : 3 : 0	14928	1	24	14953	81	66	15, 15	Surplus
111	Rg	282		C_{22}	4 : 3 : 1	14958	1	28	14987	81	66	15, 15	Surplus
112	Cn	285		C_{22}	4 : 4 : 0	15124	1	32	15157	81	66	15, 15	Surplus
113	Nh	286		C_{22}	5 : 4 : 1	15154	1	36	15191	81	66	15, 15	Surplus
114	Fl	289		C_{22}	5 : 5 : 0	15320	1	40	15361	84	66	18, 18	Surplus
115	Mc	290		C_{22}	6 : 5 : 1	15350	1	44	15395	84	66	18, 18	Surplus
116	Lv	293		C_{22}	6 : 6 : 0	15516	1	48	15565	84	66	18, 18	Surplus
117	Ts	294		C_{22}	7 : 6 : 1	15546	1	52	15599	84	66	18, 18	Surplus
118	Og	294		C_{22}	7 : 7 : 0	15508	1	56	15565	84	66	18, 18	Surplus
119	–	297		C_{22}	8 : 7 : 1	15674	1	60	15735	69	66	3, 3	Unfilled fixture
120	–	298		C_{22}	8 : 8 : 0	15704	1	64	15769	72	66	6, 6	Unfilled fixture
121	–	301		C_{22}	9 : 8 : 1	15870	1	68	15939	75	66	9, 9	Unfilled fixture
122	–	302		C_{22}	9 : 9 : 0	15900	1	72	15973	78	66	12, 12	Unfilled fixture
123	–	305		C_{22}	10 : 9 : 1	16066	1	76	16143	81	66	15, 15	Unfilled fixture
124	–	306		C_{22}	10 : 10 : 0	16096	1	80	16177	84	66	18, 18	Unfilled fixture
125	–	309		C_{22}	11 : 10 : 1	16262	1	84	16347	87	66	21, 21	Unfilled fixture
126	–	310		C_{22}	11 : 11 : 0	16292	1	88	16381	90	66	24, 24	Unfilled fixture

References

[1] Bureau International des Poids et Mesures. Si base unit: second. Accessed 26 October 2026. `https://www.bipm.org/en/si-base-units/second`.

[2] Consultative Committee for Time and Frequency. Mise en pratique for the definition of the second in the si, 2019. BIPM, SI Brochure Appendix 2. `https://www.bipm.org/documents/20126/41489667/SI-App2-second.pdf/3c76fec8-04d9-f484-5c3c-a2e280a0f248`.

[3] Helmut Hellwig, Robert F. C. Vessot, Martin W. Levine, Paul W. Zitzewitz, David W. Allan, and David J. Glaze. Measurement of the unperturbed hydrogen hyperfine transition frequency. *IEEE Transactions on Instrumentation and Measurement*, IM-19(4):200–209, 1970. Printed p. 200: one-sigma uncertainty 0.002 Hz. `https://tf.nist.gov/general/pdf/13.pdf`.

[4] Joint Committee for Guides in Metrology. Evaluation of measurement data: Guide to the expression of uncertainty in measurement, 2008. JCGM 100:2008. https://www.bipm.org/documents/20126/2071204/JCGM_100_2008_E.pdf.

[5] Joint Committee for Guides in Metrology. Guide to the expression of uncertainty in measurement: Part 6, developing and using measurement models, 2020. JCGM GUM-6:2020. https://www.bipm.org/documents/20126/2071204/JCGM_GUM_6_2020.pdf.

[6] F. G. Kondev, M. Wang, W. J. Huang, S. Naimi, and G. Audi. The nubase2020 evaluation of nuclear physics properties. *Chinese Physics C*, 45:030001, 2021. https://www-nds.iaea.org/amdc/ame2020/nubase_4.mas20.txt.

[7] H. S. Margolis, G. Panfilo, F. Collini, M. Pizzocaro, A. D. Ludlow, R. Siadat, T. Ido, and S. Bize. Data associated with the 2025 adjustment of standard frequencies, 2026. Raw input, output, covariance, README and analysis notes. https://www.bipm.org/en/doi/10.59161/StdFreq2026.

[8] National Institute of Standards and Technology. Beams of atoms: The first atomic clocks. Accessed 26 October 2026. https://www.nist.gov/atomic-clocks/how-atomic-clocks-work/beams-atoms-first-atomic-clocks.

[9] National Institute of Standards and Technology. Bohr magneton in hz/t: 2022 codata recommended value. Accessed 26 October 2026. https://physics.nist.gov/cuu/Constants/Table/allascii.txt.

[10] National Institute of Standards and Technology. Handbook of basic atomic spectroscopic data: Atomic data for hydrogen. Accessed 26 October 2026. https://physics.nist.gov/PhysRefData/Handbook/Tables/hydrogentable1_a.htm.

[11] National Institute of Standards and Technology. Atomic spectra database (ASD) and ground levels and ionization energies for the neutral atoms, 2025. NIST Physical Measurement Laboratory, https://physics.nist.gov/PhysRefData/ASD/.

[12] National Nuclear Data Center. Evaluated nuclear structure data file: Adopted levels and experimental updates, 2026. Source snapshot and exact record locators in the Matter I isotope catalog; retrieved 1 November 2026. https://www.nndc.bnl.gov/ensdf/.

[13] Reginald Poyau. Axiomatic fundamentalism (AF): A logical protocol for traceable research (all versions), 2025. All versions. DOI: https://doi.org/10.5281/zenodo.17561186.

[14] Reginald Poyau. Axiomatic fundamentalism calculus (AFC), v2.4, 2025. Authority binding: v2.4, source commit 21e66cd46b9a69a1996ed4fcecda53d831b414f8. Concept DOI: https://doi.org/10.5281/zenodo.17795590.

[15] Reginald Poyau. Alpha↔omega dynamics (AΩD): The hidden temporal dynamics of stokes; the art of the leprechaun: Fractal calculus – $\mathfrak{S}$ (v41.02, main and manual), 2026. Released v41.02; source commit 6ac87ad518652249fed67af491ea4aab9d7225ce. DOI: https://doi.org/10.5281/zenodo.23426642.

[16] J. E. Sansonetti and W. C. Martin. Handbook of basic atomic spectroscopic data. *Journal of Physical and Chemical Reference Data*, 34(4), 2005. Caesium data, printed p. 1652. https://www.nist.gov/document/jpcrd690pdf.

[17] Daniel A. Steck. Cesium d line data, 2025. Revision 2.3.4, section 3. https://steck.us/alkalidata/cesiumnumbers.pdf.

[18] M. Wang, W. J. Huang, F. G. Kondev, G. Audi, and S. Naimi. The AME 2020 atomic mass evaluation (II). *Chinese Physics C*, 45:030003, 2021.

[19] M. Wang, W. J. Huang, F. G. Kondev, G. Audi, and S. Naimi. The ame 2020 atomic mass evaluation (ii). tables, graphs and references. *Chinese Physics C*, 45:030003, 2021. `https://www-nds.iaea.org/amdc/ame2020/mass_1.mas20.txt`.